DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II
WEEK 11: LECTURE NOTE

Differentiation and Integration of Logarithmic and Exponential Functions

The Natural Logarithmic Function: Integration

Definition 1 (**Definition of the Natural Logarithmic Function**). *The natural logarithmic function is defined by*

$$\ln x = \int_1^x \frac{1}{t}dt, \quad x > 0.$$

The domain of the natural logarithmic function is the set of all positive real numbers.

Recall the number
$$e = 2.71828182846$$

Definition 2 (**Definition of** e). *The letter* e *denotes the positive real number such that*

$$\ln e = \int_1^e \frac{1}{t}dt = 1.$$

Recall the Properties of the Natural Logarithmic Function discussed in MAT111 and Lesson 7 of this course. It follows from the definition that

$$\frac{d}{dx}[\ln x] = \frac{1}{x}, \quad x > 0.$$

Theorem 3 (**Log Rule for Integration**). *Let* u *be a differentiable function of* x.

1. $\displaystyle\int \frac{1}{x}dx = \ln|x| + C$
2. $\displaystyle\int \frac{1}{u}du = \ln|u| + C$

Using the Log Rule for Integration

1. $\displaystyle\int \frac{2}{x}dx = 2\ln|x| + C = \ln x^2 + C$

2. $\displaystyle\int \frac{1}{4x-1}dx = \frac{1}{4}\int \overbrace{\frac{1}{4x\text{-}1}}^{\frac{1}{u}}\ \overbrace{(4)dx}^{du} = \frac{1}{4}\int \frac{1}{u}du = \frac{1}{4}\ln|u| + C = \frac{1}{4}\ln|4x-1| + C$

3. $\displaystyle\int_0^3 \frac{x}{x^2+1}dx = \frac{1}{2}\int_0^3 \overbrace{\frac{1}{x^2+1}}^{\frac{1}{u}}\ \overbrace{(2x)dx}^{du} = \frac{1}{2}\int_1^{10} \frac{1}{u}du = \frac{1}{2}[\ln|u|]_1^{10} = \frac{1}{2}[\ln 10 - \ln 1] = \frac{1}{2}\ln 10$

4. $\displaystyle\int \frac{3x^2+1}{x^3+x}dx = \int \overbrace{\frac{1}{x^3+x}}^{\frac{1}{u}} \overbrace{(3x^2+1)dx}^{du} = \int \frac{1}{u}du = \ln|u| + C = \ln|x^3+x| + C$

5. $\displaystyle\int \frac{\sec^2 x}{\tan x}dx = \int \overbrace{\frac{1}{\tan x}}^{\frac{1}{u}} \overbrace{(\sec^2 x)dx}^{du} = \int \frac{1}{u}du = \ln|u| + C = \ln|\tan x| + C$

6. $\displaystyle\int \frac{x^2+x+1}{x^2+1}dx = \int [1 + \frac{x}{x^2+1}]dx = \int 1dx + \int \frac{x}{x^2+1}dx = x + \frac{1}{2}\ln(x^2+1) + c =$ $x + \ln\sqrt{x^2+1} + C$

GUIDELINES FOR INTEGRATION

1. Learn a basic list of integration formulas.
2. Find an integration formula that resembles all or part of the integrand, and, by trial and error, find a choice of u that will make the integrand conform to the formula.
3. When you cannot find a u-substitution that works, try altering the integrand. You might try a trigonometric identity, multiplication and division by the same quantity, addition and subtraction of the same quantity, or long division. Be creative.
4. If you have access to computer software that will find antiderivatives symbolically, use it.

The Natural Exponential Function: Integration

Recall the following definition.

Definition 4 (Definition of the Natural Exponential Function). *The inverse function of the natural logarithmic function $f(x) = \ln x$ is called **the natural exponential function** and is denoted by*

$$f^{-1}(x) = e^x.$$

That is,

$$y = e^x \quad \text{if and only if} \quad x = \ln y.$$

Note that

$$(1)\quad \ln e^x = x, \quad x \in \mathbb{R}, \quad \text{and}$$
$$(2)\quad e^{\ln x} = x, \quad x > 0.$$

By the Chain Rule

$$1 = \frac{d}{dx}[x] = \frac{d}{dx}[\ln e^x] = \frac{1}{e^x}\frac{d}{dx}[e^x].$$

So that

$$\boxed{\frac{d}{dx}[e^x] = e^x}$$

and hence

$$\boxed{\int e^x dx = e^x + C}.$$

Example 5. *Integrating Exponential Functions*

1. $\displaystyle\int \frac{e^{\frac{1}{x}}}{x^2}dx$
2. $\displaystyle\int \sin x \ e^{\cos x}dx$
3. $\displaystyle\int_0^1 e^{-x}dx$
4. $\displaystyle\int_0^1 \frac{e^x}{1+e^x}dx$